

EEG Stroke Decision Support System

Report generated: 2026-09-27 20:23

Patient #1

Known Clinical Status: Stroke

PREDICTION RESULT

Stroke

Probability: 70.1% | Risk: HIGH

EEG FEATURES

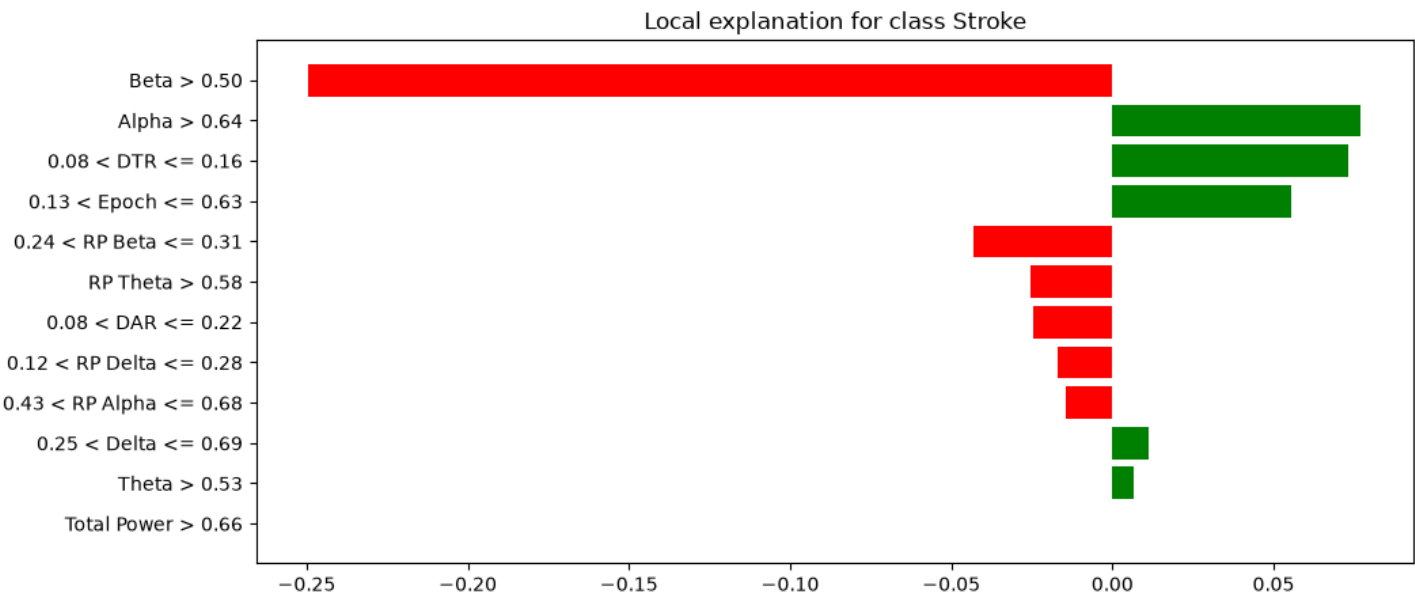
Feature	Raw Value	Normalized [0,1]
Delta	3.0660	0.5953
Theta	3.4990	0.6790
Alpha	2.3240	0.8234
Beta	1.0730	0.5098
RP Delta	0.3078	0.2403
RP Theta	0.3512	0.6117
RP Alpha	0.2333	0.4799
RP Beta	0.1077	0.2868
DTR	0.8800	0.0811
DAR	1.3193	0.1200
Total Power	9.9620	0.7568
Epoch	18.0000	0.3023

AI EXPLANATION (LIME)

Feature Condition	Impact	Direction
Beta > 0.50	0.2493	Control +
Alpha > 0.64	0.0772	Stroke +
0.08 < DTR <= 0.16	0.0731	Stroke +
0.13 < Epoch <= 0.63	0.0556	Stroke +
0.24 < RP Beta <= 0.31	0.0429	Control +
RP Theta > 0.58	0.0254	Control +
0.08 < DAR <= 0.22	0.0247	Control +
0.12 < RP Delta <= 0.28	0.0169	Control +
0.43 < RP Alpha <= 0.68	0.0145	Control +
0.25 < Delta <= 0.69	0.0113	Stroke +
Theta > 0.53	0.0066	Stroke +
Total Power > 0.66	0.0001	Control +

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DIAGNOSTIC RECOMMENDATIONS

Urgency Level: URGENT

Recommended Action: Immediate comprehensive stroke workup recommended

Recommended Tests (6):

Magnetic Resonance Imaging (MRI)

Brain imaging to detect stroke location, type, and extent

Complete Blood Count (CBC)

Blood cell counts to detect infection, anemia, or clotting disorders

Coagulation Profile (PT/INR, aPTT)

Blood clotting function to guide anticoagulation therapy

Basic Metabolic Panel (BMP)

Electrolytes, glucose, kidney function assessment

D-Dimer Test

Fibrin degradation product to detect active blood clotting

Lipid Profile

Cholesterol and triglyceride levels for cardiovascular risk

CLINICAL REASONING

- EEG-based prediction: Stroke (probability: 70.1%)
- Risk classification: HIGH

MODEL INFORMATION

Architecture:	Ensemble Echo State Network (E-ESN)
Ensemble Size:	7 ESN models with soft voting
Reservoir Size:	200 neurons per ESN
Validation:	Leave-One-Out Cross-Validation (LOOCV)
Explainability:	KernelSHAP (global) + LIME (local)
Reference:	Bouazizi & Ltifi (2024), Decision Support Systems 178